

Increasing the Robustness of the Fine-tuned Multilingual Machine-Generated Text Detectors

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Abstract

Since the proliferation of LLMs, there have been concerns about their misuse for harmful content creation and dissemination. Recent studies justify such fears, providing evidence of LLM vulnerabilities and high misuse potential. It has been shown that humans struggle to reliably distinguish high-quality machine-generated text from authentic human-written text. Therefore, it is crucial to develop automated means to accurately detect machine-generated content. While multiple detectors of machine-generated texts have been researched so far, a known issue is their low generalizability to unseen generators, domains or languages. This work¹ addresses the problem by proposing a robustness-oriented LLM training methodology for the detection task, making the detectors more robust against obfuscation and more generalizable to out-of-distribution data. The fixed-size ablation study confirms that the gains are not simply a function of more data.

1 Introduction

The number, variety, enhanced capabilities, and simple usage of large language models (LLMs) cause growing concerns about a massive generation of harmful content with LLMs being potentially maliciously misused (Borji, 2023; Zhuo et al., 2023; Crothers et al., 2023). Previous research has demonstrated the ability of LLMs to produce disinformation texts (Vykopal et al., 2024; Lucas et al., 2023), even personalized ones (Zugecova et al., 2025; Gabriel et al., 2024), frauds for academic exams (Achiam et al., 2023), or plagiarism (Wahle et al., 2022). The fears of misuse are more severe due to indistinguishability between human-written texts and high-quality machine-generated texts (MGTs) (Sadasivan et al., 2025). Thus, the MGT detection (MGTD) belongs to the key chal-

¹The source code available at: https://github.com/kinit-sk/robust_mgt_detector

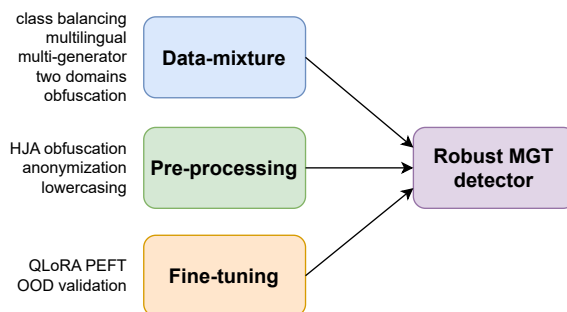


Figure 1: Overview of the proposed robustness-oriented MGT detector training methodology.

lenges associated with state-of-the-art LLMs as identified by Kaddour et al., 2023.

Since the harmful online content, such as misinformation, disinformation, propaganda, or impersonation, can be also generated or adjusted by LLMs, its automated detection can crucially help to fight such phenomena. Indication that a potentially harmful content is generated by using artificial intelligence (AI) can alleviate its eventual effect on people. By its statistical nature, the AI generated content should be considered less credible (in comparison to authentic human-written texts), mainly due to the simplicity and speed of creating the content as well as a potential presence of hallucinations (unintended misleading information).

However, the state-of-the-art MGTD methods suffer from performance degradation in out-of-distribution (OOD) settings, such as new generators, domains, languages, and paraphrasing or adversarial attacks (Antoun et al., 2023; Li et al., 2024). Therefore, increasing their robustness in this manner is the primary concern. Ensuring that some data are really OOD (in relation to MGTD training data) is quite challenging since even languages are related to each other differently (language families, writing scripts), the same regarding generators (architectural similarities, training data), and other aspects as well. However, there exist

measures of similarities between data distributions, such as MAUVE score (Pillutla et al., 2021).

This work addresses this problem by mixing the training data in a way to include multiple languages, domains, generators and even obfuscated (attack) data, along with multiple unique pre-processing steps (see Figure 1). We specifically target these research questions:

RQ1 *Can multilingual MGT detectors be made more robust to OOD data through diverse fine-tuning data mixtures?*

RQ2 *How do different foundation model architectures and scales behave under robust MGTD fine-tuning methodology?*

To the best of our knowledge, this is the first attempt to systematically fine-tune robust multilingual MGT detectors. Although such detectors manifest a small performance degradation on in-distribution data, the OOD performance is significantly increased, making them more suitable to be deployed in the real-world settings.

The key contributions of this work include:

- **A systematic methodology for robust fine-tuning** for MGT detection task, using diverse data for training and OOD data for validation, including unique pre-processing steps.
- **Comparison of variety of foundational models** robustly fine-tuned as MGT detectors by evaluation on in-distribution as well as OOD datasets.
- **Controlled evidence that gains are not a pure data-volume artifact**, where fixed-size experiments (32k/100k/1M) and a component-wise ablation (v0–v6) isolate which elements of the methodology drive OOD robustness gains.
- **Introduction of highly diverse mixture dataset** (MIX), along with its 2k-samples subset for validation (MIX2k), covering 7 languages, over 10 domains, and over 50 generators with variable generation settings. We provide the data construction source code for replication purpose (see Appendix B).

2 Related Work

The evaluation of detection robustness is unfortunately focused mostly on a single aspect in existing

works. For example, MULTITuDE (Macko et al., 2023) was focused on cross-lingual generalization, MAGE (Li et al., 2024) on cross-domain generalization, and RADAR (Hu et al., 2023) or OUTFOX (Koike et al., 2024) on adversarial robustness.

There are attempts to combine multiple aspects in the evaluation, such as (Dugan et al., 2024; Macko et al., 2024). However, the combined evaluation on completely unrelated OOD data (different generators, languages, domains, generation settings) is still lacking (especially in multilingual scenario). SemEval-2024 Task 8 multilingual dataset (Wang et al., 2024) provides a way to test a sort of multi-generator, multi-domain, and multilingual performance. However, due to inconsistent inclusion of various data (e.g., some domains represented only in a single language), the evaluation might be somewhat biased. Therefore, one should train the detectors on another data and use this dataset only for OOD evaluation.

In comparison to existing works, our approach combines their strengths and eliminates their weaknesses. Specifically, our approach integrates multiple evaluation dimensions considered separately in prior work while enabling unified training and evaluation.

3 Robustness-oriented Detector Training

A natural option for increasing OOD performance is to include all available data in the training. However, it would result in a problem of leaving no data for OOD evaluation. We cope with the problem by combining a wide set of diverse datasets into the MIX evaluation dataset that is used for OOD evaluation. We leave only carefully selected data (unrelated to evaluation data) for training and show how our unique strategy (overviewed in Figure 1 and below) empirically improves the detection robustness.

Data-mixture. Previous works have shown good cross-lingual capabilities (even to unseen languages) of MGT detectors when trained on high amount of languages. Therefore, we cover **the widest set of languages** for the training (over 44). For better generalization of detectors to unseen domains, we must include at least **two domains** in the training for the same generators and languages, to prevent biases to specific pairs. Therefore, we combine social-media texts and news articles in the training (as having distinct writing styles), leaving other domains for OOD evaluation. And finally,

to increase adversarial robustness of detectors, we include **obfuscated texts** (i.e., modified by adversarial attack, paraphrasing, and backtranslation to evade the detection) in the training. More details about the datasets can be found in Section 3.1.

Pre-processing. To further increase the robustness of the detection towards obfuscation, we introduce a combined **HomoglyphJoinerAttack** (see Section 3.2) that is applied on-the-fly to modify a portion of the training data and thus make the detector resistant to homoglyphs (visually similar characters that might confuse humans) and invisible spaces. We further apply **anonymization** and **lower-casing** to the texts, in order the detector to focus on the meaning rather than on visual appearance (and minimize biases towards personal information). More details regarding pre-processing can be found in Section 3.2.

Fine-tuning. For training, we use the LLM fine-tuning for the sequence classification task. Based on the model size, we either use half-precision full fine-tuning, or we use **QLoRA** parameter-efficient fine-tuning approach. We manually tune the hyper-parameters for all tested model-variant to be able to successfully learn the task. We tend to use the same hyper-parameters for all the models for comparison purposes. More details about the process and the used hyper-parameters can be found in Section 3.3. During the fine-tuning process, a small portion of **OOD data is used for validation** – i.e., the selection of model checkpoint that is the most generalizable to OOD.

3.1 Datasets

For training and evaluation purposes, we used a combination of multiple datasets described below.

MultiSocial (Macko et al., 2025) A dataset of real social-media texts in 22 languages from 5 platforms, containing also the generated counterparts (by using 3 iterations of paraphrasing) by 7 generators. We have balanced the train split of this dataset by additional human samples (using the published procedure²), covering also languages out of the intended 22 (specifically, 40 languages with at least 1,000 samples).

MULTITuDE_v2 (Macko et al., 2024) A subset of MULTITuDE_v2 dataset of obfuscated MULTITuDE (Macko et al., 2023) texts, covering the three

selected authorship obfuscation methods (DFT-Fooler adversarial attack, ChatGPT paraphrasing, and m2m100-1.2B backtranslation). Due to modification (obfuscation) of the texts by machines, we include only machine-labeled samples (uncertain labeling of machine-obfuscated human texts).

MULTITuDE_v3 (Macko et al., 2025) An extended version of the MULTITuDE (Macko et al., 2023) dataset of real human-written news articles in 22 languages accompanied by counterparts generated by 7 generators, created for OOD evaluation by (Macko et al., 2025). We have balanced their train split of the dataset by additional human samples (using the same published procedure³).

SemEval (Wang et al., 2024) Multilingual dataset used during SemEval-2024 Task 8.

MIX Our dataset created by merging, de-duplicating and sampling 100k human and 100k machine samples from 18 existing datasets for binary MGTD task (for more details about the dataset creation process, see Appendix B). It represents the most diverse dataset of this kind created so far.

MIX2k A subset of a MIX dataset, covering various domains, 7 languages and over 50 generators. It contains 1k human and 1k machine samples.

For *training (fine-tuning)* of detectors, we combine the train splits of the first three datasets (Table 1). After de-duplication, the obtained set includes 427,195 human samples and 554,216 machine samples out of 2 domains (news and social media), 44 languages (of more than 1,000 samples), and 16 generators. A small imbalance in human vs machine classes was addressed by up-sampling of the minority class (i.e., human) during the training. The MIX2k dataset (containing mostly OOD data) was used for validation during the fine-tuning, i.e., for checkpoint selection.

For the *in-distribution evaluation*, we use the test splits of the MULTITuDE_v3 and MultiSocial datasets. For the *OOD evaluation*, we use the test

³<https://github.com/kinit-sk/mgt-detection-benchmark>

Dataset	Human	Machine	Generator	Language
MultiSocial	290,618	290,618	7	40
MULTITuDE_v2	0	120,090	7	3
MULTITuDE_v3	136,577	143,527	7	22
MIX2k	1,000	1,000	>50	7

Table 1: Text samples counts for the train splits of the included datasets (MIX2k used for validation only).

²<https://github.com/kinit-sk/multisocial>

Dataset	Human	Machine	Generator	Language
MultiSocial	17,367	121,460	7	22
MULTITuDE_v3	6,502	44,566	7	22
MIX	99,759	100,000	>50	7
SemEval	20,238	22,140	7	4

Table 2: Text samples counts for the test splits of the evaluation datasets.

split of the SemEval, as well as the full MIX dataset. The statistics regarding sample counts for testing are summarized in Table 2.

3.1.1 Data Distribution Evaluation

To measure data distribution similarities (mostly in regard to OOD evaluation), we are calculating MAUVE (Pillutla et al., 2021) scores and Frontier integrals (Liu et al., 2021) between individual train and test sets. MAUVE uses calculation of Kullback–Leibler (KL) divergences between the two data distributions in a quantized embedding space of a foundation model (we employ BERT-base-multilingual-cased). Frontier integral is a statistical summary of divergence frontiers, measuring the quality-diversity trade-off. Baseline train set represents the training data used for baseline fine-tuned detector. Robust train set is the proposed data mixture used for robust detector fine-tuning. For MAUVE, higher values indicate that the distributions are closer to each other. For Frontier integral, the same is indicated by lower values. The results are provided in Table 3. It can be clearly seen that the MultiSocial and MULTITuDE_v3 test splits are closer to training data distributions than the MIX and SemEval test datasets.

3.2 Texts Pre-processing

To increase the robustness against obfuscation methods (preventing the detection), we modify a portion of machine-generated training samples by our new modified attack, combining a generic HomoglyphAttack (introduced by Macko et al., 2024) with insertion of zero-width-joiner (ZWJ) character

Train	Test	MAUVE	Frontier integral
baseline	MultiSocial	0.6493	0.1449
robust	MultiSocial	0.3985	0.2420
baseline	MULTITuDE_v3	0.2131	0.3536
robust	MULTITuDE_v3	0.1852	0.3780
baseline	MIX	0.0268	0.6971
robust	MIX	0.0367	0.6461
baseline	Semeval	0.0071	0.9099
robust	Semeval	0.0086	0.8792

Table 3: Evaluation of data distribution shift between train and test datasets.

```
def homoglyphJoiner(input_text):
    output = ""
    for char in input_text:
        if (random.random() < H_PROBABILITY):
            hcount = len(confusable_characters(char))
            idx = int(random.random()*hcount)
            output += confusable_characters(char)[idx]
        else:
            output += char
        if (random.random() < ZWJ_PROBABILITY):
            output += "\u200D"
    return output
```

Figure 2: HomoglyphJoinerAttack algorithm.

(inspired from GPTZeroBypasser⁴).

This attack, called HomoglyphJoinerAttack (HJA), uses the whole confusables table⁵ to pseudo-randomly replace letters for their homoglyphs (visually similar characters). Also, it pseudo-randomly inserts a character that is invisible for humans (non-printable). A detailed Python algorithm is provided in Figure 2. We use both, the H_PROBABILITY (a probability of a character to be replaced by its random homoglyph) and ZWJ_PROBABILITY (a probability of insertion of the ZWJ character after the current character of the string) of 0.05. In this way, we have modified about 10% of MGTs of the training set.

In order for the detector to focus on the meaning of the text rather than on appearance and to avoid biases towards specific personal information, we have algorithmically anonymized the texts by replacing the identified email addresses, user mentions, and phone numbers by the corresponding tags of [EMAIL], [USER], [PHONE]. The specific regular expressions we are using to identify such information are available in the enclosed source code. Afterwards, the texts are converted to lowercase as part of the pre-processing. This kind of anonymization and lowercase conversion is done for each text, not only for training but for testing as well.

3.3 Fine-tuning Procedure

We used two kinds of fine-tuning, based on the model size. For smaller models, we employ the full fine-tuning process utilizing half precision, with the default AdamW optimizer with learning rate of $2e-6$, batch size of 32, validation each 2,000 steps, running at max for 10 epochs or 4 days. For

⁴<https://github.com/jayyt12161/GPTZero-Bypasser/blob/main/GPTZero-Bypasser.py>
⁵<https://www.unicode.org/Public/security/8.0.0/confusables.txt>

the models having more than 2B parameters, we employ QLoRA (Dettmers et al., 2024) parameter efficient fine-tuning process with the weighted cross entropy with logits for loss calculations and 4-bit quantization using BitsAndBytes⁶. The LoRA configuration⁷ used an α of 16, a dropout of 0.1, r of 64, and the task type of sequence classification targeting all linear modules. In comparison to full fine-tuning, the QLoRA fine-tuning version used paged AdamW optimizer with learning rate of $2e - 5$, gradient accumulation of 4 steps, and evaluation each 200 steps.

We used these hyper-parameter values for all models to make them comparable (i.e., robust variants use the exact same optimization hyper-parameters as baseline). However, if a smaller batch size was necessary due to memory constraints in case of bigger models, we tuned the values accordingly.

4 Experimental Settings

As a baseline, we use the **mDeBERTa-v3-base** model (as shown by existing works as the best of smaller multilingually pre-trained models, e.g., by Macko et al., 2023) fine-tuned on a combination of MULTITuDE_v3 and MultiSocial datasets. Specifically, for baseline training, we use their train splits without our balanced extension by additional human samples, i.e., about 486k samples with about 61k being human written (up-sampling of minority class is used for fine-tuning), using 10% of train data for validation. We selected 5 models with diverse architectures and numbers of parameters for robust fine-tuning: 1) mDeBERTa-v3-base (He et al., 2022) to compare the effect of the robust fine-tuning with the same underlying model used as a baseline; 2) Qwen2 (Yang et al., 2024) models with 0.5B and 1.5B parameters to see the effect of model size within the same family; 3) Yi-1.5-6B (AI et al., 2024) to see the effect of bilingually pre-trained model; and 4) Gemma-2-9b-it (Team, 2024) to see the effect of a bigger, instruction fine-tuned model.

As the evaluation metric, we are primarily using **AUC ROC** (area under curve of the receiver operating characteristic) as a classification-threshold independent metric reflecting generic detection capability. For a fine-grained comparison, we are also using **TPR@5%FPR**, representing true positive rate (TPR) using the classification threshold

tuned to reach 5% of false positive rate (FPR) based on ROC curve, and Macro F1 (macro average of the F1 score) reflecting a balance between precision and recall metrics.

5 Results

When evaluating on in-distribution data (Table 4), we can notice the highest performance of the baseline detector in both AUC ROC and TPR @ 5% FPR metrics in news as well as social-media texts. This indicates that the baseline is a strong in-distribution model. However, we hypothesize that such a baseline, trained purely on the train splits of the evaluation datasets, suffers from over-fitting to the data distribution and thus generalizes worse to OOD data.

All of the detectors are performing better on the MULTITuDE_v3 news articles than on the Multisocial social-media texts, where the lowest performance is achieved by robustly fine-tuned mDeBERTa-v3-base and Qwen2-1.5B models.

When looking at the per-language AUC ROC performance (Table 6), we can see that the detection works well for all the languages evenly, with a small exception of surprise languages (ga, gd, sl) in MultiSocial (i.e., not enough social-media texts for training in some languages). This further **evokes a lower generalizability to OOD data**.

As the results for OOD evaluation indicate (Table 5), the baseline detector is the worst performing on both OOD datasets, which **confirms our hypothesis about over-fitting**. The AUC ROC

Detector	MULTITuDE_v3		MultiSocial	
	AUC ROC	TPR@5%FPR	AUC ROC	TPR@5%FPR
Gemma-2-9b-it	0.9914	0.9798	0.9563	0.8400
Qwen2-0.5B	0.9785	0.9316	0.9582	0.8413
Qwen2-1.5B	0.9883	0.9460	0.9549	0.7781
Yi-1.5-6B	0.9748	0.9115	0.9474	0.8008
mDeBERTa	0.9959	0.9797	0.9540	0.7750
baseline	0.9944	0.9875	0.9746	0.8862

Table 4: Evaluation of detection performance on in-distribution data. The highest value in each column is boldfaced.

Detector	MIX		SemEval	
	AUC ROC	TPR@5%FPR	AUC ROC	TPR@5%FPR
Gemma-2-9b-it	0.8901	0.5227	0.9448	0.8284
Qwen2-0.5B	0.6499	0.0000	0.8434	0.0000
Qwen2-1.5B	0.7588	0.3064	0.9391	0.8287
Yi-1.5-6B	0.8167	0.0000	0.8922	0.0000
mDeBERTa	0.6669	0.0502	0.8666	0.6843
baseline	0.5502	0.0000	0.8305	0.0000

Table 5: Evaluation of detection performance on out-of-distribution data. The highest value in each column is boldfaced.

⁶<https://pypi.org/project/bitsandbytes>

⁷<https://pypi.org/project/peft>

Detector	Test Language [AUC ROC]																			
	ar	bg	ca	cs	de	el	en	es	et	ga	gd	hr	hu	nl	pl	pt	ro	ru	sk	sl
MULTITuDE_v3	Gemma-2-9b-it	0.999	0.998	0.988	1.000	0.995	0.997	0.997	0.990	0.959	0.994	0.921	0.996	0.996	0.991	1.000	0.998	0.998	0.995	0.998
	Qwen2-0.5B	0.958	0.985	0.992	0.995	0.989	0.936	0.980	0.982	0.953	0.974	0.941	0.983	0.975	0.992	0.989	0.996	0.993	0.966	0.992
	Qwen2-1.5B	0.993	0.988	0.997	0.997	0.987	0.925	0.994	0.994	0.963	0.990	0.966	0.991	0.985	0.997	0.994	0.996	0.994	0.985	0.998
	Yi-1.5-6B	0.969	0.967	0.991	0.991	0.984	0.919	0.985	0.980	0.951	0.983	0.941	0.982	0.958	0.993	0.982	0.984	0.983	0.959	0.990
	mDeBERTa	0.998	0.998	0.998	1.000	0.996	0.992	0.991	0.994	0.994	0.992	0.990	0.998	0.999	0.999	0.997	0.997	0.997	0.989	0.999
MultiSocial	baseline	0.996	0.996	0.996	1.000	0.990	0.991	0.980	0.991	0.993	0.997	0.992	0.999	0.999	0.996	0.997	0.994	0.996	0.992	0.993
	Gemma-2-9b-it	0.959	0.985	0.954	0.977	0.957	0.936	0.975	0.969	0.936	0.886	0.811	0.976	0.981	0.958	0.973	0.962	0.974	0.968	0.965
	Qwen2-0.5B	0.955	0.985	0.963	0.979	0.953	0.956	0.972	0.967	0.974	0.858	0.822	0.968	0.984	0.956	0.960	0.964	0.950	0.963	0.960
	Qwen2-1.5B	0.967	0.984	0.965	0.976	0.958	0.937	0.977	0.971	0.923	0.857	0.828	0.959	0.956	0.964	0.965	0.967	0.943	0.962	0.960
	Yi-1.5-6B	0.934	0.975	0.956	0.968	0.945	0.931	0.967	0.963	0.966	0.883	0.842	0.960	0.971	0.952	0.958	0.965	0.924	0.951	0.923
MultiSocial	mDeBERTa	0.973	0.989	0.943	0.964	0.948	0.961	0.964	0.954	0.971	0.838	0.773	0.957	0.982	0.951	0.957	0.951	0.949	0.956	0.969
	baseline	0.967	0.991	0.978	0.984	0.975	0.975	0.981	0.979	0.984	0.919	0.869	0.978	0.988	0.973	0.981	0.974	0.974	0.970	0.971

Table 6: Per-language AUC ROC performance of the detectors on in-distribution data. The highest value in each dataset per each language is boldfaced.

Detector	Threshold	MultiSocial			MULTITuDE_v3			MIX			SemEval			→ Average Macro F1
		FNR	FPR	Macro F1	FNR	FPR	Macro F1	FNR	FPR	Macro F1	FNR	FPR	Macro F1	
Gemma-2-9b-it	0.500	0.157	0.052	0.767	0.026	0.038	0.942	0.141	0.222	0.818	0.128	0.088	0.891	0.855
Qwen2-0.5B	0.990	0.116	0.071	0.806	0.043	0.076	0.903	0.396	0.374	0.615	0.249	0.133	0.806	0.782
Qwen2-1.5B	0.900	0.101	0.109	0.813	0.068	0.039	0.877	0.299	0.353	0.674	0.187	0.038	0.884	0.812
Yi-1.5-6B	0.900	0.127	0.092	0.788	0.093	0.048	0.841	0.176	0.305	0.758	0.151	0.170	0.840	0.807
mDeBERTa	0.970	0.130	0.043	0.798	0.033	0.022	0.935	0.489	0.296	0.604	0.256	0.133	0.803	0.785
baseline	0.999	0.169	0.030	0.761	0.036	0.018	0.932	0.520	0.402	0.537	0.076	0.375	0.774	0.751

Table 7: Estimation of generic detection performance using the tuned threshold for each detector (for deployment). The best achieved value of each metric is boldfaced (the lowest for FNR and FPR, the highest for Macro F1). The Gemma-2-9b-it detector is offering the best overall detection performance.

differences between baseline and the robustly fine-tuned mDeBERTa-v3-base **are statistically significant**, confirmed by two-sided DeLong test (DeLong et al., 1988) on both MIX and SemEval OOD datasets (p-values < 0.001). The better performance of robustly fine-tuned detectors using our augmented datasets for training clearly **shows the benefits of our proposal**.

The mDeBERTa-v3-base model (the same base model as the baseline) was able to achieve by 21% higher AUC ROC performance on the MIX dataset and by 4% on the SemEval dataset. Regarding the TPR@5%FPR metric, the results show that the baseline detector was not able to achieve such a FPR for any classification threshold. On the other hand, the robustly fine-tuned version was able to detect almost 70% of MGTs on the SemEval dataset for 5% FPR. It must be noted that not only the baseline, but also the Yi-1.5-6B and Qwen2-0.5B detectors have not detected any MGT in the 5% FPR conditions. Overall, the best detector on OOD data is the Gemma-2-9b-it robustly fine-tuned model, detecting 52% and 83% of MGTs of the MIX and SemEval datasets for the 5% FPR conditions.

A deeper analysis using the ROC curves of the detectors for individual datasets (Appendix C) revealed that the used OOD data are really challenging for them, while the in-distribution data are resulting in almost perfect classification capabilities. Figure 5 shows that the performance of the base-

line detector is only negligibly better than a random guessing (a diagonal) using the MIX dataset.

5.1 Application in the Real World

Based on such in-distribution and OOD evaluation, we performed a manual classification-threshold tuning to achieve optimal trade-off detection (i.e., the highest Macro F1 across the datasets). **Thresholds are selected to maximize deployment utility rather than to evaluate the proposed robustness-oriented training methodology**, and all robustness claims rest on the AUC ROC threshold-independent metric in the main results above. The tuned thresholds are needed for application of the detectors in the real world (i.e., in the wild, without known labels) to make the decision and to approximate the actual expected performance (e.g., FPR). The thresholds are tuned to balance the performance in each dataset individually and to maximize the performance across all the datasets – i.e., to identify the optimal threshold to be applied in the real world. The results for the tuned thresholds are provided in Table 7, where we provide the FNR, FPR, and Macro F1 metrics for each dataset, as well as the average of Macro F1 across the four datasets. The results indicate the significantly better performance of the Gemma-2-9b-it detector compared to the others, representing also the largest model. The Qwen2-1.5B detector offers also good performance, while being much smaller

than the best model (9B vs 1.5B parameters $\sim 6\times$ smaller); thus, offering a good trade-off between the performance and inference costs.

5.2 Comparison to Existing Detectors

Table 8 summarizes the detection performance results on each of the used datasets achieved by existing best performing statistical detection methods, applicable in zero-shot manner (without training), namely Binoculars (Hans et al., 2024), Fast-DetectGPT (Bao et al., 2023), and LLM-Deviation (Wu and Xiang, 2023). Macro F1 is calculated based on manual threshold tuning, analogously to the results in Table 7. Similarly, Table 9 shows the performance of existing pre-trained detection methods, fine-tuned on different datasets. BLOOMz represents BLOOMZ-3B-mixed-detector (Sivesind and Winje, 2023), Longformer refers to Longformer Detector (Li et al., 2024), and OpenAI represents RoBERTa-Large OpenAI Detector (Solaiman et al., 2019).

The comparison to the existing methods indi-

Dataset	Metric	Binoculars	Fast-DetectGPT	LLM-Deviation
MultiSocial	AUC ROC	0.7753	0.7717	0.7454
	TPR@5%FPR	0.3069	0.3063	0.1561
	Macro F1	0.4433	0.3450	0.3406
MTD_v3	AUC ROC	0.8169	0.8184	0.6666
	TPR@5%FPR	0.5647	0.5576	0.0829
	Macro F1	0.6510	0.6328	0.5928
MIX	AUC ROC	0.5369	0.5147	0.5053
	TPR@5%FPR	0.2347	0.2321	0.0928
	Macro F1	0.5552	0.5504	0.5089
SemEval	AUC ROC	0.8782	0.8835	0.9098
	TPR@5%FPR	0.7353	0.7087	0.2437
	Macro F1	0.8207	0.8161	0.8628
↓ Average Macro F1		0.6176	0.5861	0.5763

Table 8: Evaluation of detection performance of zero-shot statistical detection methods. MTD_v3 denotes MULTITuDE_v3 dataset. The highest value in each row is boldfaced.

Dataset	Metric	BLOOMz	Longformer	OpenAI
MultiSocial	AUC ROC	0.7523	0.4609	0.6802
	TPR@5%FPR	0.2233	0.0377	0.1963
	Macro F1	0.5715	0.3924	0.4750
MTD_v3	AUC ROC	0.5473	0.6082	0.5493
	TPR@5%FPR	0.0914	0.1057	0.1362
	Macro F1	0.3654	0.4862	0.4753
MIX	AUC ROC	0.5625	0.7274	0.3997
	TPR@5%FPR	0.0000	0.2383	0.0641
	Macro F1	0.5138	0.6527	0.4228
SemEval	AUC ROC	0.7452	0.9177	0.6498
	TPR@5%FPR	0.2967	0.5663	0.1616
	Macro F1	0.6189	0.8059	0.5611
↓ Average Macro F1		0.5174	0.5843	0.4835

Table 9: Evaluation of detection performance of existing fine-tuned detection methods. MTD_v3 denotes MULTITuDE_v3 dataset. The highest value in each row is boldfaced.

cates that **even our worst performing fine-tuned detector performs better (based on Macro F1 average) than the usually applied publicly available detectors (statistical or pretrained).**

5.3 Adversarial Robustness

For evaluation of adversarial robustness, we use obfuscated MGTs of MULTITuDE_v2 test set in addition to their unobfuscated human samples. These data include obfuscation by 10 authorship obfuscation methods (Macko et al., 2024), covering also methods that have not been used for adversarial training of our detectors. In our evaluation, we use 8 of these 10 obfuscation methods, since the original authors have disqualified DIPPER and Pegasus-paraphrase as unusable in multilingual settings (caused severe damages to the text readability).

The results in Table 10 indicate that **the robustly fine-tuned detectors show strong resilience to these kinds of obfuscations** (AUC ROC values are even slightly higher on obfuscated data than on original data), which otherwise cause evasion of detection. On the other hand, the baseline mDeBERTa-v3-base version has decreased performance in all the cases (up to -27% in case of HomoglyphAttack) except for a slight increase on the ChatGPT paraphrased data. In all the cases, the robustly fine-tuned mDeBERTa achieves higher AUC ROC values than the baseline version.

The MULTITuDE_v2 data are considered in-distribution, since a part of their train set has been combined into the robust training of our detectors. Therefore, a slightly higher performance of mDeBERTa-v3-base in comparison to other robustly fine-tuned detectors is not unexpected (analogously to the performance on MULTITuDE_v3 in Table 4).

Obfuscator	Gemma-2-9b-it	Qwen2-1.5B	mDeBERTa	baseline
original	0.9757	0.9719	0.9887	0.9821
adversarial				
ALISON	0.9790	0.9744	0.9872	0.9804
DFTFooler	0.9834	0.9783	0.9946	0.9776
backtranslation				
m2m100-1.2B	0.9842	0.9840	0.9918	0.9780
nllb-200-dist 1.3B	0.9843	0.9878	0.9913	0.9819
paraphrasing				
ChatGPT	0.9816	0.9905	0.9953	0.9926
text edits				
HomoglyphAttack	0.9860	0.9850	0.9986	0.7213
GPTZeroBypasser	0.9880	1.0000	0.9987	0.8505
GPTZzss	0.9806	0.9776	0.9917	0.9793

Table 10: Evaluation of adversarial robustness (AUC ROC comparison) of the selected detection methods. The highest value in each row is boldfaced.

6 Ablation Study

To find out the effect of individual components of the proposed robustness-oriented fine-tuning methodology, we executed an ablation study using the mDeBERTa model. For this purpose, we iteratively added individual components, starting with the baseline (v0) and resulting with robustly fine-tuned version (v6) reported in Table 5. For evaluation and comparison of these intermediate versions, we use the OOD datasets (MIX and SemEval) and the AUC ROC metric.

The results are provided in Table 11. It can be seen that balancing the training data using additional human-written texts helps significantly, especially in case of the SemEval dataset. After the inclusion of obfuscated machine-generated data into training, the performance on the MIX dataset is boosted significantly; however, it drops on the SemEval dataset. When switching from in-distribution validation to OOD validation data, the performance is unchanged. HJA modification of a portion of machine-generated data slightly increased the performance on the SemEval data. After using the final text preprocessing (anonymization + lowercasing) of training as well as testing data, v6 achieves peak performance on the MIX dataset, while it is one of the best on the SemEval data. v3 also shows that adding 1% of MIX data (MIX2k) for validation **has not caused an overfit to the test distribution of the MIX dataset**. To rule out the data-distribution leakage effect completely (due to using MIX2k for OOD validation), we trained another detector (v5) by avoiding OOD validation during robust fine-tuning. It thus corresponds to the v2 + HJA + anonymization version. We can see that the performance is very close to the final robust v6 version, significantly better than the baseline in OOD evaluation (also confirmed by two-sided DeLong tests with p-values < 0.001).

To also rule out data size effect (due to training dataset extension), we executed an experiment of using a fixed number of train samples in both, baseline and robust training. The training is stopped after using 1M train samples and the last model checkpoint is selected, i.e., the checkpoint selection is invariant to the validation-set performance. We can observe (the bottom part of Table 11) an increased performance of the baseline on both OOD datasets (indicating the previous overfit to train distribution based on in-distribution validation), while the SemEval performance achieves the top. How-

Detector	MIX	SemEval
v0 (baseline)	0.5502	0.8305
v1 (baseline + extended data)	0.5824	0.9144
v2 (v1 + obfuscated data)	0.6302	0.8617
v3 (v2 + OOD validation)	0.6336	0.8553
v4 (v3 + HJA)	0.6318	0.8718
v5 (v2 + HJA + anonymization)	0.6497	0.8770
*v6 (v4 + anonymization)	0.6513	0.8679
Size effect (fixed number of seen train samples)		
v0 (1M)	0.6067	0.9178
v6 (1M)	0.6291	0.8281
v0 (100k)	0.6087	0.8248
v6 (100k)	0.6134	0.7072
v0 (32k)	0.6160	0.7121
v6 (32k)	0.6184	0.6266

Table 11: Effect of individual components of robust fine-tuning on detection performance (AUC ROC) on out-of-distribution data. v6 is the robustly fine-tuned mDeBERTa-v3-base (marked by *). The highest value in each column is boldfaced.

ever, the robust fine-tuning still achieves a significantly better performance than the baseline on the MIX dataset (two-sided DeLong tests with p-values < 0.001). It shows that **the boosted performance of the proposed training methodology is not simply because of more data**, indicating an important role of checkpoint selection during fine-tuning. Also, to simultaneously eliminate the effect of the minority class (highly imbalanced for baseline, with 60k human samples available), we use stricter limits of 100k and 32k train samples. The results show consistently increased performance of robustly trained detector on the MIX dataset, with the difference being statistically significant (DeLong tests showing the highest p-value of 0.0033 in case of 32k fixed train samples, still below the standard 0.05). These findings demonstrate that **class imbalance in the training data is not the primary driver of degraded OOD performance, thereby validating the necessity of the proposed multifactor data diversification strategy**.

7 Conclusions

The fine-tuned MGTD models are known to generalize poorly to out-of-distribution (OOD) data. In this work, we augmented training data to better balance the number of samples of machine and human classes. Besides, we included obfuscated data directly into the detection model training; thus potentially making it more resistant against obfuscation as well. The results showed that such robustly fine-tuned detectors offer lower false positive rates and

the proposed training methodology is able to boost performance (AUC ROC) by up to 21%. OOD robustness in multilingual MGTD depends more on training distribution diversity and OOD-aware model selection than on architecture alone. The performance of different detection models varies, providing an option to select the best suited one giving the trade-off between the detection performance and inference time (or availability of GPU accelerators). The further work is required to optimize the hyper-parameters of the fine-tuning process to boost the detection performance of the best detection models even higher while offering even better generalizability to OOD data. Future work could also explore learned or adaptive obfuscation-detection strategies that generalize beyond the specific perturbation patterns used in this work.

Limitations

Although we have tried to cover variety of domains, generators, and languages, the study is still limited in out-of-distribution evaluation. There is much more space to explore, and new text generation LLMs appear each month that could cause further performance degradation. Our work improves robustness to currently known distribution shifts, but it does not solve generalization against future unseen generators permanently. We have also done a manual hyper-parameters optimization of fine-tuning process; however, we have covered just a small set of options, where a further tuning might increase the generalizability of the detectors.

Ethical Considerations

Our work is focused on making the detection methods more robust to out-of-distribution data. We use the existing datasets in our work in accordance with their intended use and licenses (for research purpose only). As a part of our work, we are not re-sharing any existing data or publishing any new dataset. For the research replicability and validation purposes, we are publishing the pre-processing, training, and evaluation source codes (for research purposes only). We have not used AI assistants in our work except for text polishing.

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A Computational Resources

For experiments regarding model fine-tuning and inference processes, we have used 1x A100 64GB GPU, cumulatively consuming around 1,000 GPU-core hours. For data augmentation, pre-processing, combining, and analysis of the results, we have used Jupyter Lab running on 4 CPU cores, without the GPU acceleration.

Regarding the environment, we have used Python 3.11.6 with key libraries of pytorch 2.6.0 (with CUDA 11.8), transformers 4.57.6, accelerate 1.12.0, bitsandbytes 0.48.2, and peft 0.18.1.

B MIX Dataset

For the diverse mixture dataset of existing labeled datasets for out-of-distribution evaluation, we have combined the samples of data (up to 5,000 text samples per language/source and label) of the following datasets: TuringBench (Uchendu et al., 2021), Grover (Zellers et al., 2019), GPT-2 (Radford et al.,

2019), GPT-3 (Brown et al., 2020), Paired (Pu et al., 2022), CNN/DailyMail (See et al., 2017), NeuralNews (Tan et al., 2020), InTheWild (Pu et al., 2023), Wikitext (Merity et al., 2016), WritingPrompts (Fan et al., 2018), CoAuthor (Lee et al., 2022), TweepFake (Fagni et al., 2021), WELFake (Verma et al., 2021), AuTexTification (Sarvazyan et al., 2023), MTG (Chen et al., 2022), MGTBench (He et al., 2024), RuATD (Shamardina et al., 2022), SynSciPass (Rosati, 2022).

The data have been combined in a common way, providing texts, labels, source, and if available also the language and domain/topic information. Afterwards, the duplicates of the texts have been removed and the final dataset has been pseudo-randomly sampled to include 100k samples for the human and machine classes. For the MIX2k validation dataset, the 1k samples of each class have been pseudo-randomly sampled. It resulted into diverse data of at least (since some are unspecified in the original datasets) 7 languages (cn, de, en, es, fr, ru, zh), 10 domains (news, tweets, unspecified social-media texts, comments, reviews, wiki texts, stories, literature, scientific articles, question answering, commonsense reasoning), 50 generators (sizes of under 1B parameters to over 100B, including pure generators, but also translators, co-writing agents, as well as unknown reddit bots), and various generation settings (e.g., diverse approaches, at least 7 different sampling strategies).

Due to various licensing of the data of the used datasets (some of which are also composites of multiple data sources), we are not re-sharing the dataset itself. However, for the research purposes (validation and replication), we are providing the data construction source code.

C Supplemental Results

In Figure 3–6, ROC curves of the detectors for individual datasets are provided.

In Figure 7 and 8, we provide a deeper analysis of the performance of the best detector of Gemma-2-9b-it on the most challenging MIX dataset. The sample counts are not balanced across the provided categories; therefore, the results are just indicative. There is a significant drop in performance for German and French languages, which are included only for machine-generated samples originated from machine translation (MTG dataset). The low performance is also observed in the commonsense-reasoning, wikitexts, and literature domains.

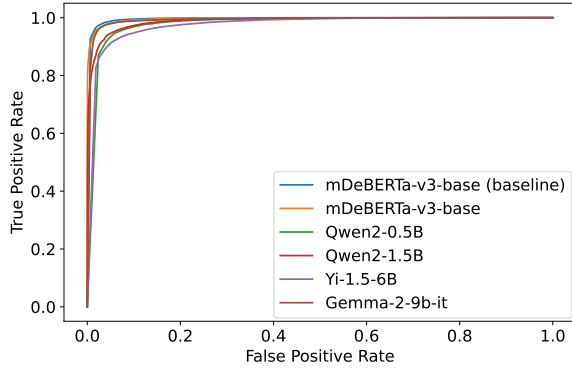


Figure 3: Receiver operating characteristic curves of the detectors using the MULTITuDE_v3 dataset.

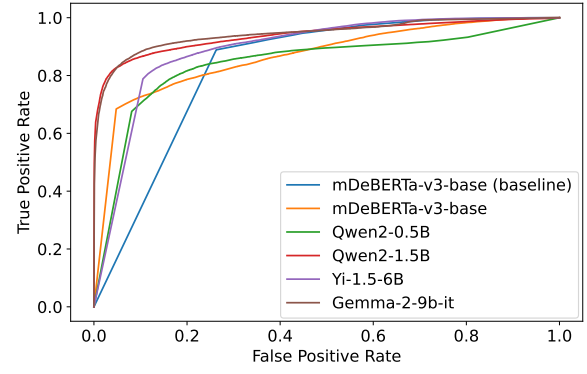


Figure 6: Receiver operating characteristic curves of the detectors using the SemEval dataset.

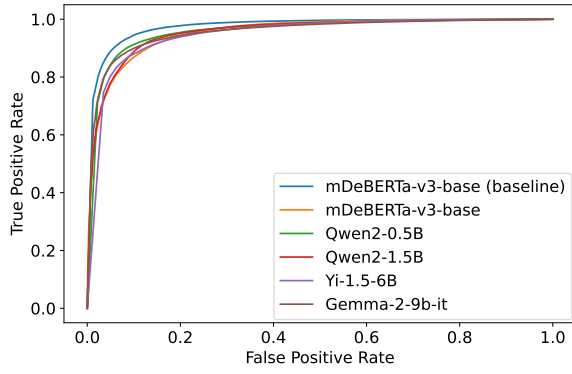


Figure 4: Receiver operating characteristic curves of the detectors using the MultiSocial dataset.

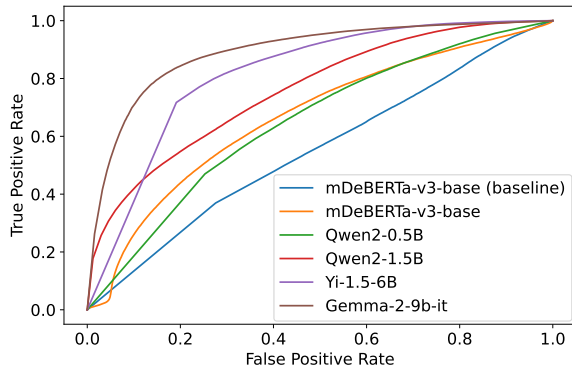


Figure 5: Receiver operating characteristic curves of the detectors using the MIX dataset.

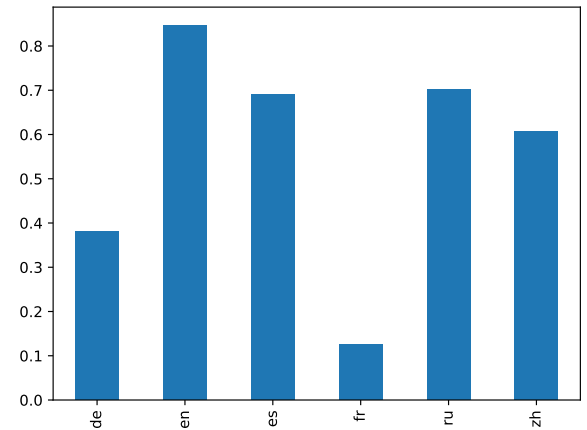


Figure 7: Per-language performance (Macro F1) of the best Gemma-2-9b-it detector.

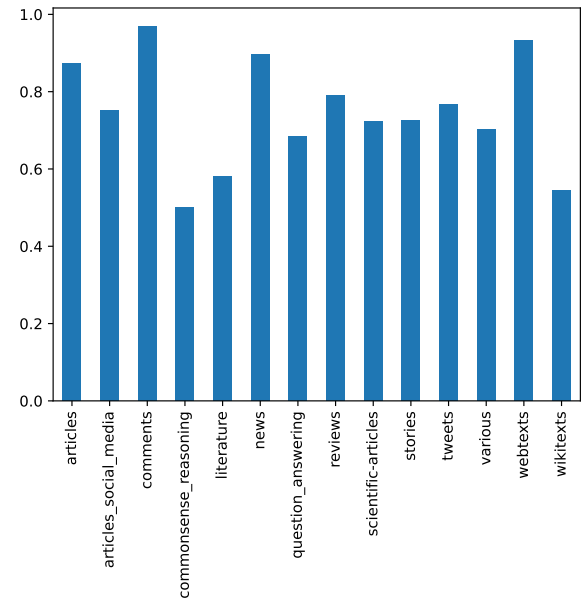


Figure 8: Per-domain performance (Macro F1) of the best Gemma-2-9b-it detector.